



DPUSGETexturePylon

Creates a dynamic video texture in the virtual world. The texture contains a live video image that is captured from a Basler Pylon compatible camera.

Note: This DPU requires that the Pylon runtime, version 4.x, is installed. For new setups, it is recommended to use DPUSGETexturePylon5 instead.

1. Dynamic parameters:

<i>ParamFile</i>	File name of a parameter file which is read when the simulation is prepared. The camera is configured according to this parameter file. If a relative path is given, the file is searched below the SILAB\lib directory. Default: (<i>empty string</i>) which means that the camera is not reconfigured when the simulation is prepared.
<i>SaveParam</i>	When the value changes from 0 to 1, the current camera settings are saved to the parameter file specified in <i>ParamFile</i> . Default: 0
<i>AutoGain</i>	Enable (1) / disable (0) the camera's automatic gain feature. Default: -1 which means the feature is left "as before"
<i>AutoWhiteBalance</i>	Enable (1) / disable (0) the camera's automatic white balance feature. Default: -1 which means the feature is left "as before"
<i>AutoExposure</i>	Enable (1) / disable (0) the camera's automatic exposure time feature. Default: -1 which means the feature is left "as before"
<i>AutoExposureTimeMinUS</i>	Minimum exposure time (in microseconds) used by the automatic exposure feature. (1 .. 1000000) Default: -1 which means the setting is left "as before"
<i>AutoExposureTimeMaxUS</i>	Maximum exposure time (in microseconds) used by the automatic exposure feature. (1 .. 1000000) Default: -1 which means the setting is left "as before" Note: If the exposure time is increased above 10000 us (= 10 ms) then the available frame rate will be decreased and the image may become unsharp.
<i>AutoGainMin</i>	Minimum gain parameter used by the automatic gain feature. (0 .. 100, recommended: 0) Default: -1 which means the setting is left "as before"
<i>AutoGainMax</i>	Maximum gain parameter used by the automatic gain feature. (1 .. 100, recommended: as low as possible) Default: -1 which means the setting is left "as before"



AutoAvgBrightness Target average brightness of the image, used by the camera's automatic gain and exposure features. (50 .. 150)
Default: -1 which means the setting is left "as before"

2. Static parameters:

View Name of the SGE view for which the video texture is up-dated. Additionally, it will be visible in all views that share textures with the specified SGE view.
Default: *SGEView*

TextureName Name of the created video texture, which must start with a dollar sign (\$).
Default: *\$Video*

CameraName Name of the camera to capture. This is the "Device User ID" that can be entered in the *Basler IP Configurator* application.
Default: (*empty string*) which means the first connected camera is used.

PixelFormat Selected pixel format of the camera. The supported values depend on the camera model, however the following values will likely work:
Mono8, Mono12, BayerRG8, BayerRG12, YUV422Packed
Default: (*empty string*) which means the camera's default startup pixel format is used.

Decimation Can be used to reduce the camera resolution. The horizontal and vertical resolution will be reduced by the specified integer number. If the visible output image is much smaller than the camera's native resolution, reducing the resolution improves the performance a lot without reducing image quality.
Default: -1 which means the camera's default resolution is used; possible values are: 1 (full resolution) ... 32

Width, Height Size (width and height in pixels) of the area of interest.
Default: -1 which means the camera's default size is used

OffsetX Distance from the left side of the sensor to the left side of the output image, in pixels. Default: -1 (camera default)

OffsetY Distance from the top side of the sensor to the top side of the output image, in pixels. Default: -1 (camera default)

3. Functional principle:

Using *DPUSGETexturePylon*, a dynamic video texture is created in the virtual world. The texture contains a live video image that is captured from a Basler Pylon compatible camera. (Both USB3 and GigE cameras should work.)

The *TextureName* parameter defines a name under which the video texture will be available in the SGE world. The name must start with a dollar sign (\$). The video texture can be used in any SGE object loaded after this DPU by using this texture name instead of specifying a texture file on the hard disk. Alternatively, *DPUSGEVideoObject* can be used to create a simple overlay object which displays the video texture.



4. Example of a SGE object which references a video texture:

```
define SIG3DObject VideoObj
{
    AlphaTest = 0.2;
    Blend = false;
    // [...]
    Texture = $MyVideo;
    TextureFilter = NoCompression | ClampS | ClampT | MagLinear |
                  MinLinear | Tex-ture;
    // [...]
};
```

5. Dependencies to other DPUs:

Please make sure that the computer on which **DPUSGETexturePylon** runs also executes **DPUSGE-World**, **DPUSGEObjectXXX** and **DPUSGEView**. The Index values of these DPUs must be set in the following order:

DPUSGEWorld.Index < DPUSGETexturePylon.Index < DPUSGEObjectXXX.Index < DPUSGEView.Index

(DPUSGEObjectXXX stands for any DPU that uses the dynamic video texture in one of its displayed objects.)

6. Example:

This DPU works well using the following script:

```
DPUSGETexturePylon VidTexture
{
    Computer = {VIS};
    Index = 50;

    TextureName = "$MyVideo";
    # Use Bayer pixel encoding for optimum transfer speed
    PixelFormat = "BayerRG8";
};
```